Yap Ze Kai

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WORK EXPERIENCE

Health Informatics Research Lab @ NUS **Jun 2025 – Current**

Research Engineer (Full-stack)

- Built a comprehensive healthcare management portal with Next.js and TypeScript, featuring patient goal tracking, authentication systems, data visualization charts, and responsive dashboards with advanced table functionality and pagination for doctors. Integrated RESTful APIs for CRUD operations on patient goals with dynamic URL parameter management.
- Maintained a comprehensive React Native mobile application for obesity tracking, implementing features including activity monitoring, nutrition logging, sleep tracking, weight management, goal setting, and gamified quest systems with real-time Fitbit data synchronization and personalized user experiences for KKH children.
- Designed and implemented a Node.js/TypeScript backend for a hospital wellness app, covering nutrition/activity/sleep logging, Fitbit integration, gamification, and an admin analytics portal on Firebase and MongoDB.
- Collaborated in a cross-functional team using Agile/Scrum methodologies, participating in weekly sprints to ensure timely delivery of high-priority features.

Singapore Armed Forces (SAF) **Apr 2022 – Nov 2023**

Army Technician/Administrative Manager, 2nd Forward Maintenance Platoon, 9th Army Maintenance Base

- Maintained, diagnosed, & repaired Terrex & Belrex fleets for the 2nd Singapore Infantry Regiment, while providing support during field missions in Singapore & at Exercise Wallaby
- Managed administrative apparatus, maintained weapon records & assisted army regulars with their duties
- Worked with Microsoft Excel & Word extensively by manipulating spreadsheets & documents to organize technical data & safety records
- Awarded the rank of Corporal First Class (CFC) and attained 9AMB's Depot Best Soldier Award in June 2023

PROJECTS

BenchMark - GitHub **May 2025 – Jul 2025**

Developed a comprehensive AI-powered fitness companion mobile application using React Native, Expo, and TypeScript. Integrated Google Gemini AI to generate personalized workout plans based on user preferences and fitness goals. Built social features including friend connections, workout sharing, and progress tracking to gamify fitness experiences. Implemented Firebase backend for real-time data management, user authentication, and social interactions.

Neighbourly - GitHub **Mar 2025**

Developed an IOS application that fosters community engagement by connecting neighbours for assistance in day-to-day tasks. Built using Swift, SwiftUI and Supabase, with features like user authentication, real-time messaging and location-based service listings.

AI Data Explorer - GitHub **Feb 2025**

Developed an interactive web application using Streamlit and Python for AI-assisted data analysis. Enables users to upload CSV/XLSX files, preview data with Pandas, and query datasets conversationally via the OpenAI API, providing full DataFrame context for accurate AI responses.

TECHNICAL SKILLS

Coding Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, SQL
Frameworks & Libraries: React, Next.js, React Native, Node.js, MongoDB, PostgreSQL
Backend & Cloud & Tools: AWS, Docker, Supabase, Firebase, PostgreSQL, Git

EDUCATION/CERTIFICATES

National University of Singapore (NUS) **Aug 2024 – Apr 2028**
Bachelor of Computing (Honours) in Computer Science

- Cumulative GPA: 4.50
- Relevant Coursework: **CS2040S** (Data Structures & Algorithms), **CS2102** (Database Systems), **CS2109S** (AI & Machine Learning), **CS2103T** (Software Engineering)

Scrimba (Certificate) **May 2025 – Jun 2025**
Scrimba: Learn React

Scrimba (Certificate) **Jun 2025 – Jun 2025**
Scrimba: Learn TypeScript

HarvardX (Certificate) **Nov 2023 – Dec 2023**
CS50x Introduction to Computer Science

Hwa Chong Institution (HCI) **Jan 2016 – Dec 2021**
GCE 'A' Level Certificate

- RP: 86.25
- Scored A in H2 Computing, H2 Mathematics & H2 Physics

LEADERSHIP & ACHIEVEMENTS

Edusave Award for Achievement, Good Leadership & Service (EAGLES) **2021, 2019**
Given up to 10% of Singaporean pupils who demonstrate leadership qualities, offer good service to community and schools, and excel in non-academic activities.

Hwa Chong Diploma with Distinction **2021**
Awarded to students with outstanding holistic achievements in various academic & non-academic areas.

Hwa Chong Table Tennis A, B & C Division Captain **2021, 2019, 2017**
Lead and represented Hwa Chong's table tennis teams from 2016 – 2021.